

US Airline Ranking 2026 H1

Exercises on the US airline reliability data, first half of 2026: 9 airlines, built from the US Department of Transportation's Air Travel Consumer Report

Teacher edition. This version includes the answer key at the end. Do not hand it to students; a separate student sheet without answers is available.

Level: high school and university. **Subjects:** statistics, mathematics, economics, business studies.

Tools: Excel or Google Sheets. No programming needed.

Before you start

The main file ranks 9 airlines on four measures: on-time arrivals (%), cancellations (%), mishandled bags (per 100 passengers) and passengers bumped from flights (per 10,000). For on-time, higher is better. For the other three, lower is better. The composite score combines the four; check the README for exactly how. Lower composite score = better overall rank.

Files used (download links and the README are on the web page of this sheet: getkibbo.com/education/exercises/us-airline-ranking):

- us-airline-ranking-2026-h1.csv (main ranking)
- us-airline-ranking-2026-h1-yoy-comparison.csv (year-over-year)
- us-airline-ranking-2026-h1-spirit-monthly.csv (Spirit monthly)
- us-airline-ranking-README.md

Exercises

Exercise 1 — Reading the data (*basic*)

1. Which airline had the best on-time rate? Which had the highest cancellation rate?

Exercise 2 — The overall winner (*basic*)

1. Allegiant Air is ranked #1 overall. Sort the main table by on-time rate only. Where does Allegiant come? How can an airline be #1 overall without being the most punctual?

Exercise 3 — Percentage points vs. percent (*intermediate*)

- Using the year-over-year file: Spirit's cancellation rate went from 1.41% (H1 2025) to 7.47% (H1 2026).
 - By how many percentage points did it rise?
 - By what percentage did it rise?
 - Delta's cancellations rose by 1.13 points, which is +92.6%. Write one sentence a journalist could use to make this sound alarming, and one that makes it sound minor. Both should be true.

Exercise 4 — The gap with the rest of the industry (*intermediate*)

- Using the Spirit monthly file, calculate the gap between the industry average on-time rate and Spirit's on-time rate for each month, and the average gap across the five months.

Exercise 5 — Check the small print (*advanced*)

- Spirit ceased operations on May 2, 2026. Look at the May row in the Spirit monthly file. Why should you be careful about treating May the same as the other months?

Exercise 6 — Is the ranking fair? (*advanced*)

- The composite score is based on each airline's rank on each measure, not the size of the differences. Spirit's on-time rate (59.84%) is more than 11 points below the next-worst airline, but in rank terms it is just "one place lower".
 - Explain what information a rank-based score throws away.
 - The four measures are weighted equally. Passengers bumped from flights is a very rare event (per 10,000). Argue for or against giving it the same weight as on-time performance.
 - If you were building the ranking for a traveller who only cares about arriving on time, how would it change?

Connect it to real life

What are you entitled to when your flight is cancelled or badly delayed? Use the Kibbo Flight Delay Compensation Calculator and the Flight Disruption Compensation & Reimbursement Letter Generator (getkibbo.com).

Answer key

Exercise 1

- Best on-time: **Alaska Airlines, 80.51%**. Highest cancellations: **Spirit Airlines, 7.47%**.

Exercise 2

1. By on-time rate alone, Allegiant comes **6th of 9** (72.71%). It ranks #1 overall because it is best or joint-best on the other three measures: lowest cancellations (0.90%), fewest mishandled bags (0.12 per 100) and no bumped passengers (0.00 per 10,000).

Exercise 3

- a) **+6.06 percentage points.**
- b) $(7.47 - 1.41) \div 1.41 = \mathbf{+429.8\%}$.
- c) Example: “Delta’s cancellations nearly doubled” (alarming) vs. “Delta cancelled about one more flight in every hundred” (minor). Both describe the same change. This is the key lesson about points vs. percent.

Exercise 4

1. January 8.7 · February 15.7 · March 25.0 · April 16.2 · May 19.0 points. Average gap: **16.92 points.**

Exercise 5

1. Spirit stopped flying on May 2, so the May figure covers at most a day or two of operations. A rate based on so few flights is much less reliable than a full month, and it shouldn’t be compared directly with January–April.

Exercise 6

- a) It throws away the size of the differences: a tiny gap and a huge gap both count as one place.
- b) Open question. Strong answers note that equal weighting is a choice, not a fact, and that a rare event can swing the ranking out of proportion to how many passengers it affects.
- c) Ranking by on-time only: Alaska 1st, Delta 2nd, United 3rd, Southwest 4th, American 5th, Allegiant 6th, Frontier 7th, JetBlue 8th, Spirit 9th.